

SOWMIYA KUMARESAN

AI Backend Engineer

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SUMMARY

AI Backend Engineer with 1.5+ years of experience building scalable AI-powered backend systems using Python and Django. Experienced in developing LLM-powered applications with Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Model Context Protocol (MCP), vector databases, REST APIs, Celery, Redis, and WebSockets. Passionate about building production-ready AI applications, conversational AI systems, and event-driven backend architectures.

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, HTML5, CSS3
AI / LLM	Large Language Models (LLMs), Generative AI, Prompt Engineering, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Model Context Protocol (MCP), OpenAI API, Transformers, Natural Language Processing (NLP), Machine Learning, Deep Learning
Vector DBs	Qdrant, FAISS, Milvus
Backend	Python, Django, Django REST Framework (DRF), Flask, REST APIs, Django Channels, WebSockets
Databases	PostgreSQL, MySQL, SQLite, Redis
Async Systems	Celery, Redis, Event-Driven Architecture, Webhook Processing, Background Tasks
Core Skills	AI Application Development, Conversational AI, Multi-Agent Systems, Backend Architecture, API Integration, Real-time Systems
Tools	Git, GitHub, Docker, Postman, VS Code, Jupyter Notebook, Google Colab, Power BI

Experience

Python Developer <i>Thaagam Foundation, Chennai</i>	<i>Jun 2025 – Jul 2026</i>
- Developed scalable backend applications using Python, Django, Django REST Framework, PostgreSQL, Celery, and Redis for AI-powered and enterprise communication platforms. - Designed and maintained RESTful APIs, webhook integrations, and asynchronous task pipelines to support reliable, high-volume event processing. - Integrated OpenAI-powered AI workflows for conversational automation, intelligent response generation, and business process automation. - Built real-time backend services using Django Channels and WebSockets, improving system responsiveness and live communication. - Collaborated with frontend developers, QA engineers, and product teams to deliver production-ready features while optimizing API performance and database queries.	
AI Engineering Intern <i>NCubeBeacons, Bangalore</i>	<i>Oct 2024 – Mar 2025</i>
- Built data preprocessing and feature engineering pipelines using Python, Pandas, and NumPy for machine learning applications. - Assisted in developing and integrating machine learning models into backend applications through REST APIs. - Worked with NLP, data visualization, and analytics tools to support AI-driven business solutions. - Developed dashboards and reports using Matplotlib and Power BI for data analysis and decision-making.	

Projects

CommAI – AI-Powered Omnichannel Communication Platform https://comm.ai.m7corporation.com/ <i>Python, Django, Django REST Framework, PostgreSQL, Redis, Celery, OpenAI API, Django Channels, WebSockets</i>	<i>2025</i>
- Developed a scalable AI-powered multi-channel communication platform integrating WhatsApp, Instagram, Facebook Messenger, and Email into a unified conversation system. - Built webhook-driven event processing pipelines to normalize provider-specific events into a common schema and process high-volume messages asynchronously using Celery and Redis.	

- Integrated **OpenAI-powered conversational AI** for automated responses, follow-ups, conversation analysis, and intelligent human-agent handoff.
- Designed **multi-tenant backend services**, real-time messaging using Django Channels and WebSockets, media handling, delivery tracking, and conversation management.

Business Intelligence Agent – AI Analytics Platform

2026

[GitHub](#)

Python, Django, PostgreSQL, LangGraph, OpenAI, Qdrant, Celery, Pandas

- Built an AI-powered analytics platform that enables users to query structured datasets using natural language.
- Developed a multi-agent workflow using LangGraph for SQL generation, visualization, and insight generation.
- Implemented RAG with Qdrant for schema-aware retrieval and contextual AI responses.
- Designed scalable REST APIs and asynchronous report generation using Celery.

EDUCATION

B.Tech in Information Technology, Paavai Engineering College
Namakkal, — *CGPA: 8.86/10.0*

2020 – 2024

CERTIFICATIONS

- **AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents** – Udemy (Ed Donner, LGENCY)
- **Google Data Analytics Professional Certificate**